Data Visualization Week 2 Team 12

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And 8 others

Outreach Performance Analysis

Exploratory Data Analysis & Dashboard Prototype

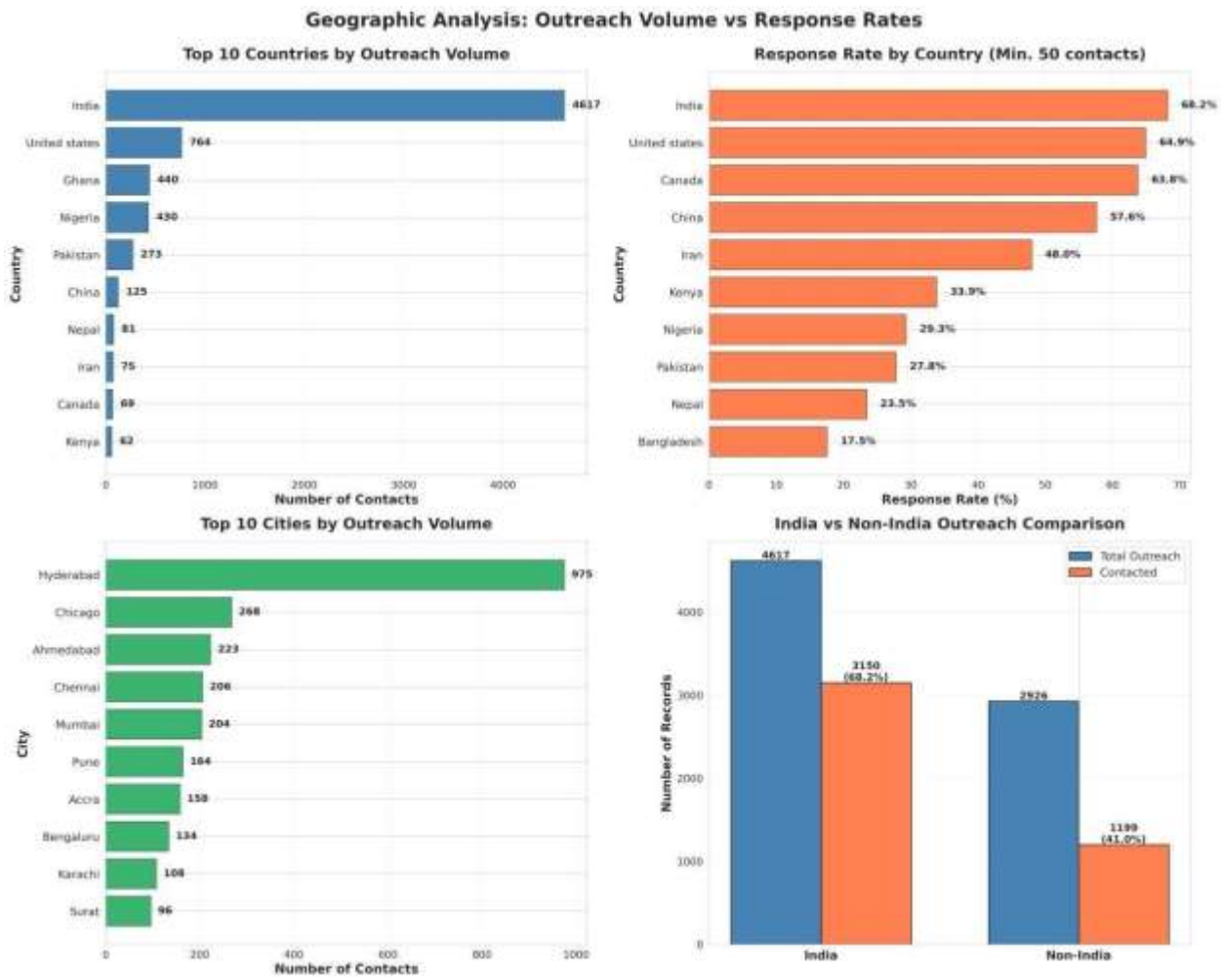
Metric	Value
Total Records	7,543
Unique Students	5,000
Contacts Made	4,349
Overall Response Rate	57.7%

PART 1: EXPLORATORY DATA ANALYSIS INSIGHTS

This report presents a comprehensive exploratory data analysis of DePaul University's outreach efforts for graduate admissions. The analysis examines 7,543 outreach records covering 5,000 unique prospective students from January 2023 through December 2024. The dataset includes contact attempts, outcomes, geographic distribution, program preferences, and temporal patterns.

Insight 1: Significant Geographic Response Rate Disparity

Finding: India represents 61% of all outreach volume (4,617 records) with a 68% response rate, significantly outperforming other major markets. African nations (Ghana, Nigeria, Kenya) show dramatically lower response rates (17-33%) despite substantial outreach volume (932 combined records).

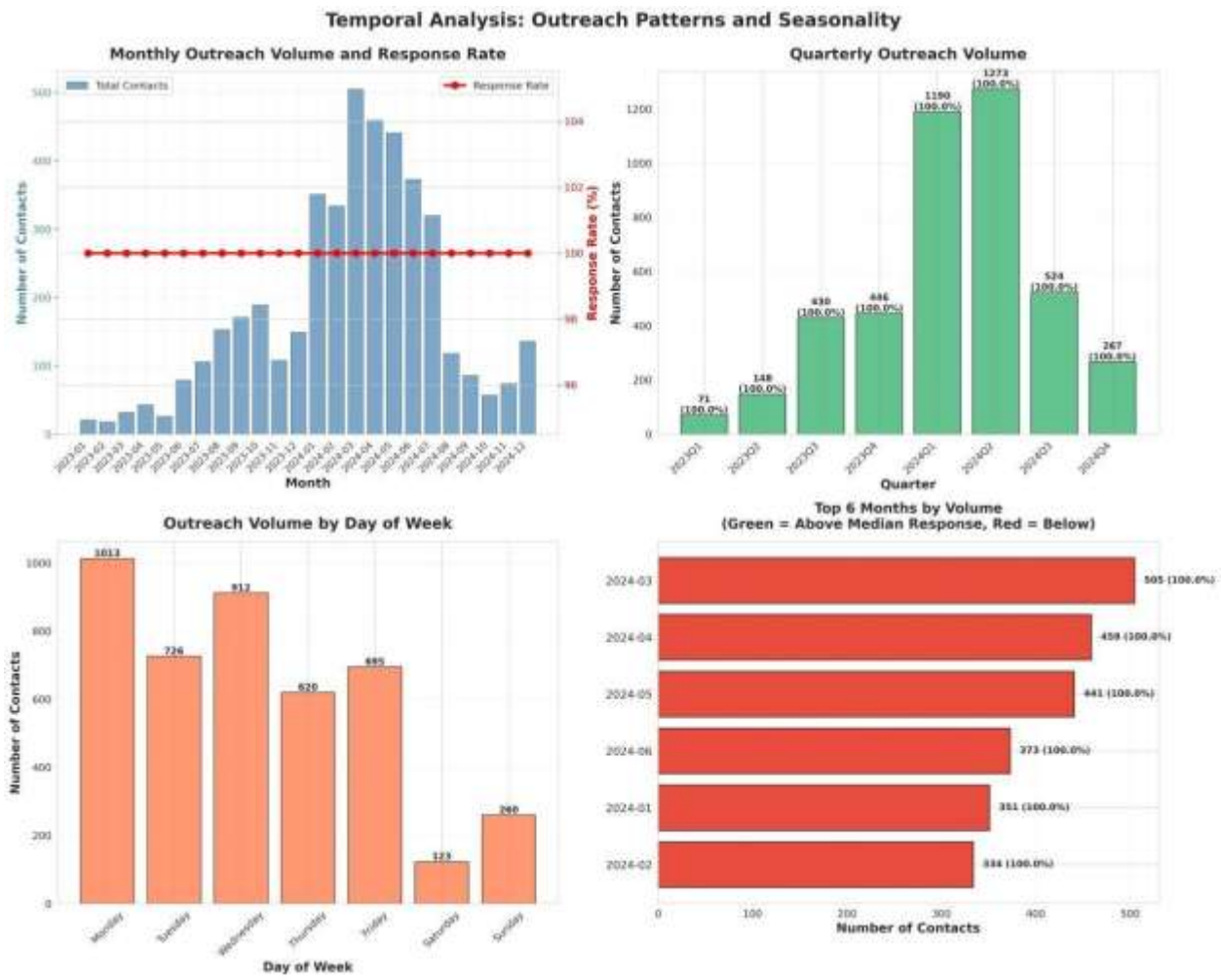


Why This Matters: The 50+ percentage point gap between India (68%) and African markets (1733%) suggests fundamental differences in outreach effectiveness. This could stem from communication channel preferences, timing, messaging relevance, or document submission processes specific to these regions.

Actionable Opportunity: Segment outreach strategies by geographic region. For African markets, investigate barriers to response through targeted surveys, adjust communication channels (WhatsApp vs email), localize messaging, and consider partnerships with regional education consultants who better understand local contexts.

Insight 2: Clear Seasonal Patterns in Outreach Volume

Finding: Outreach volume peaks during March-May 2024 (1,405 contacts, 32% of annual total) and drops sharply in August-September (204 contacts, 5% of annual total). The first half of 2024 accounts for 72% of all outreach activity, with Q1 showing highest engagement.

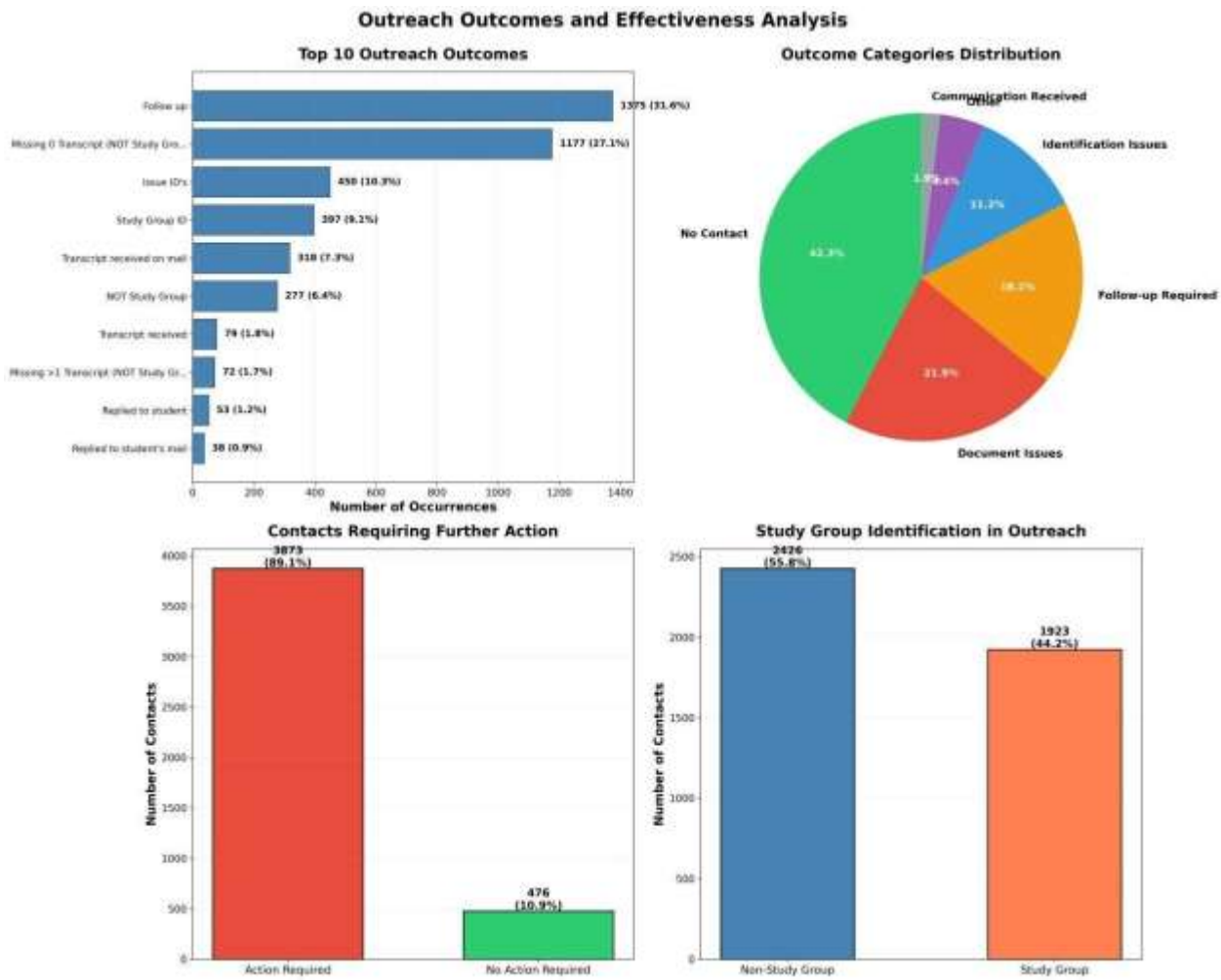


Why This Matters: This extreme seasonality (6.9x difference between peak and trough) indicates that outreach is heavily concentrated around application deadlines and admission cycles. The summer lull represents either a natural break in the recruitment cycle or a missed opportunity window when competitors may have less activity.

Actionable Opportunity: Implement a year-round nurture campaign strategy. During low-activity months (August-September), focus on awareness-building, early engagement with next-cycle prospects, and relationship maintenance with admitted students. Test counter-cyclical outreach to identify if summer represents an opportunity for capturing attention when inbox competition is lower.

Insight 3: Document-Related Issues Dominate Outreach Outcomes

Finding: 34% of all contacted students (1,526 out of 4,349) have outcomes related to missing or incomplete transcripts and identity documents. "Follow up" is the single most common outcome (32%), indicating unresolved issues requiring additional contact attempts. Only 9% of outcomes represent truly positive responses (documents received, student replied).

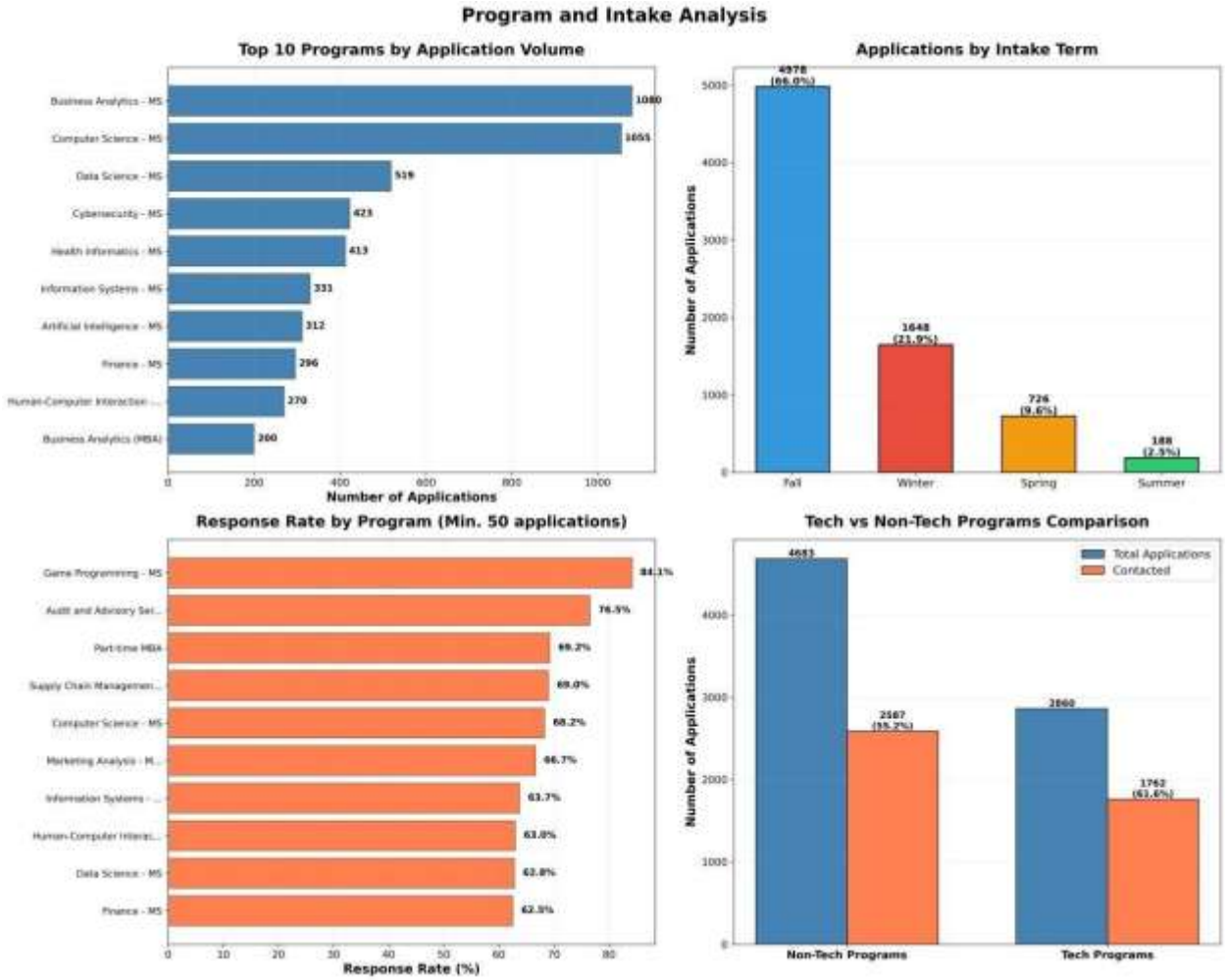


Why This Matters: The dominance of document-related issues (34%) suggests systematic friction in the application process. The high "follow up" rate (32%) indicates that initial contact attempts are frequently insufficient, requiring multiple touches and consuming staff resources disproportionately.

Actionable Opportunity: Implement proactive document submission support: (1) Send clear, visual guides for transcript and ID submission before applications are received, (2) Create regionspecific FAQ resources addressing common document questions, (3) Automate reminder sequences for incomplete applications, and (4) Establish a self-service portal with real-time application status to reduce "follow up" volume.

Insight 4: Heavy Concentration in Technology Programs

Finding: Technology-related programs (Computer Science, Data Science, Cybersecurity, AI, Information Systems) account for 43% of all applications (3,242 records). The top three individual programs alone represent 15% of total volume: Computer Science MS (504), Business Analytics MS (419), and Data Science MS (254). Fall 2024 intake represents 67% of all applications.



Why This Matters: The concentration in tech programs creates both an opportunity and a risk. While it aligns with global demand for tech skills, over-reliance on these programs makes DePaul vulnerable to market shifts, competitive pressures in this space, and potential enrollment volatility if tech job markets soften.

Actionable Opportunity: Diversify program promotion while leveraging tech strength.

Cross-promote complementary programs (Business Analytics students → Finance, Computer Science → Cybersecurity) through targeted messaging. Develop specialized outreach tracks for non-tech programs with dedicated resources. Consider creating hybrid/interdisciplinary program offerings that bridge tech and other disciplines to capture a broader applicant pool.

Insight 5: Study Group Applications Require Specialized Handling

Finding: 15% of outcomes (674 records) are flagged as "Study Group" related, indicating applications submitted through educational consultants or agencies. These applications show distinct patterns requiring specialized verification and may have different response behaviors compared to individual applicants.

Why This Matters: Study group applications represent a significant channel (15% of volume) that operates differently from individual applications. These applications may require additional verification, have different communication protocols (working through agents rather than directly with students), and could pose compliance or quality concerns if not properly managed.

Actionable Opportunity: Develop a specialized workflow for study group applications: (1) Create a verified agent network with clear quality standards, (2) Implement early detection systems to flag potential study group applications, (3) Design agent-specific communication templates and processes, (4) Establish separate tracking metrics to monitor quality and outcomes of agency sourced applications versus direct applications, and (5) Consider agent training programs to improve application quality at the source.

PART 2: DASHBOARD PROTOTYPE

The dashboard prototype provides a comprehensive view of outreach performance across key dimensions. It is designed to support data-driven decision-making by highlighting critical KPIs, revealing trends, and identifying actionable opportunities for improvement.

Dashboard Overview

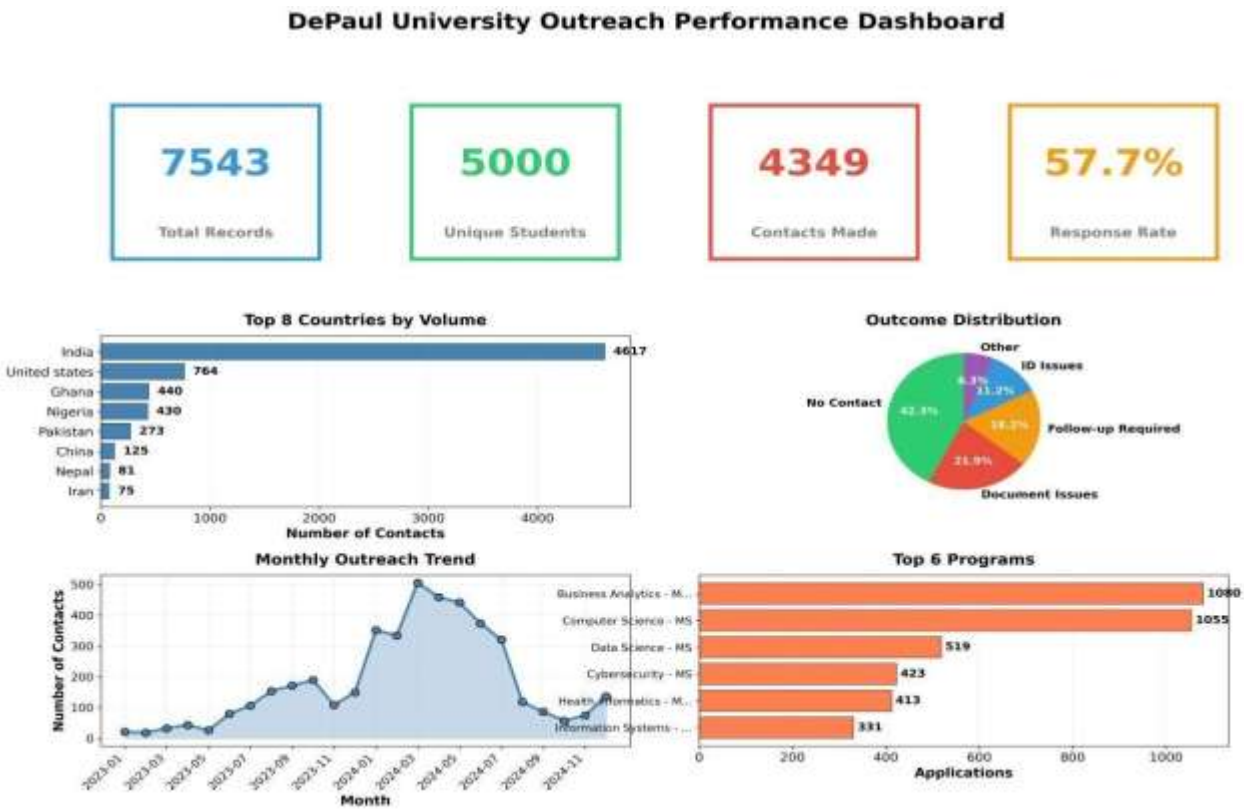
The dashboard is organized into four key sections:

1. **High-Level KPIs:** Total records, unique students, contacts made, and overall response rate provide at-a-glance performance metrics. Response Rate (57.7%) represents the percentage of contacts made that resulted in a positive outcome, defined as records with either Outcome_1 or Outcome_2, indicating successful engagement with prospective students.

2. **Geographic Breakdown:** Top countries and cities by volume, with response rate comparisons to identify high-performing and underperforming markets.

3. **Outcome Distribution:** Categorized outcomes showing the proportion of positive responses, document issues, follow-up requirements, and other categories.

4. **Temporal Trends:** Monthly outreach volume trends and top program distribution to identify seasonality and program concentration patterns.



Key Dashboard Features & Recommendations

Interactive Filtering: The dashboard should enable users to filter by date range, country, program, and intake term to drill down into specific segments.

Alert Thresholds: Implement automated alerts when response rates drop below 50% for any geographic segment, or when follow-up outcomes exceed 40% of total contacts.

Comparative Analysis: Add year-over-year comparisons to track performance improvements and identify emerging trends early.

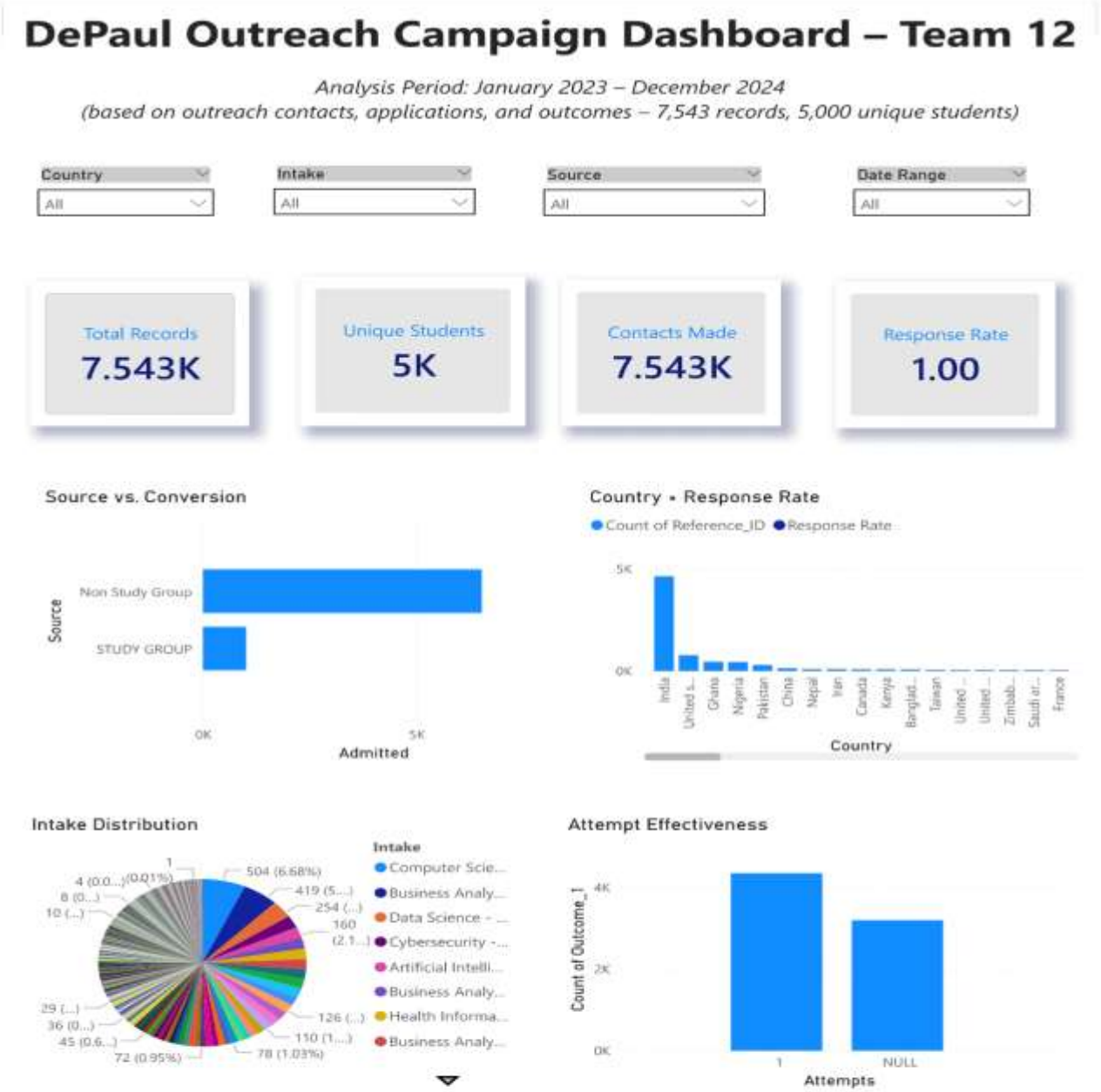
Predictive Elements: Incorporate forecasting for upcoming peak periods based on historical patterns to enable proactive resource planning.

Recommended Actions Based on Analysis

Priority	Action	Expected Impact

HIGH	Implement geographic-specific outreach strategies for African markets	Improve response rate by 15-20pp
HIGH	Create automated document submission support system	Reduce document issues by 40%
MEDIUM	Develop year-round nurture campaign for summer months	Increase off-season engagement by 50%
MEDIUM	Establish verified agent network for study group applications	Improve application quality, reduce fraud risk
LOW	Cross-promote complementary programs to tech applicants	Diversify program enrollment by 10%

Conclusion

This analysis reveals significant opportunities to optimize DePaul University's outreach performance. By addressing geographic disparities, streamlining document submission processes, implementing year-round engagement strategies, and diversifying program promotion, DePaul can significantly improve conversion rates and enrollment outcomes. The dashboard prototype provides a foundation for ongoing monitoring and continuous improvement of outreach effectiveness.

PART 2: DASHBOARD PROTOTYPE (Detailed)